

Transforming Procedural Instructions into In-Situ Augmented Reality Guides with InstructAR

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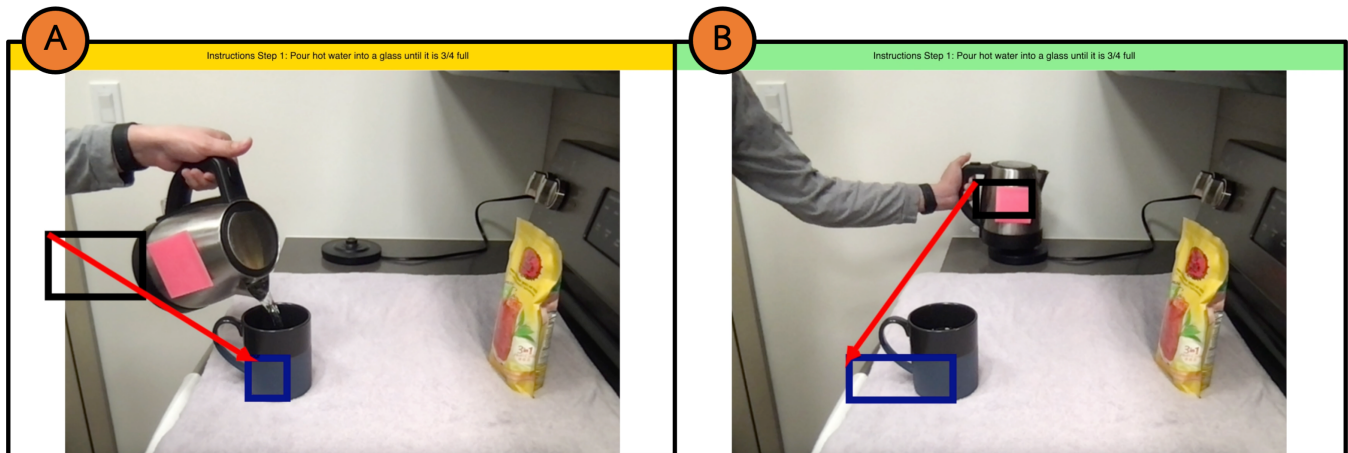


Figure 1: Screenshots of InstructAR for making a cup of tea. A) The system displays "Instruction Step 1: Pour hot water into a glass until it is 3/4 full" with an arrow guiding from the hot water kettle to the cup. B) After the user successfully completes the step, the upper label turns green to indicate successful completion of the step. The system progresses to the next step after that.

ABSTRACT

Following manual procedural instructions can often be frustrating due to the mental gap between the instructions and their real-world application. Variations in settings can further compound this confusion for those attempting to follow them. To address this problem, we propose InstructAR, a system under development that bridges textual instructions with the real world through Augmented Reality (AR) guides. With this tool, users can easily follow clear and actionable step-by-step instructions, as related objects are highlighted and AR guides indicate necessary actions. InstructAR also provides feedback upon task completion. This proof-of-concept leverages Natural Language Processing and Computer Vision techniques. Our work aims to reduce cognitive load and errors, making it easier for users to follow manual procedural instructions accurately.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality.

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UIST Adjunct '24, October 13–16, 2024, Pittsburgh, PA, USA

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ACM ISBN 979-8-4007-0718-6/24/10

<https://doi.org/10.1145/3672539.3686321>

KEYWORDS

Augmented Reality, Instruction, Computer Vision

ACM Reference Format:

Panayu Keelawat and Ryo Suzuki. 2024. Transforming Procedural Instructions into In-Situ Augmented Reality Guides with InstructAR. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3672539.3686321>

1 INTRODUCTION

Do you ever find yourself struggling with how to tackle unfamiliar procedures, relying on written instructions for guidance? This is not uncommon due to several reasons. First, task performers often struggle to identify which objects to focus on. For example, when assembling furniture, they might not know the names of specific parts or the sequence in which to use them. Second, it's often unclear what specific actions are required and on which objects. For instance, "Secure the shelf to the wall" is vague about how to precisely secure the shelf. Lastly, there's no confirmation if a step was performed correctly. In procedural tasks with multiple steps, incorrect execution of one step can cause subsequent steps to fail. While solutions such as watching a demo video can be tremendously helpful [7], there may be discrepancies between the demo setting and the actual work environment.

Textual instructions, which are widely available, have rich information already. According to research, integrating visual elements

with text can significantly enhance instruction understanding, underscoring the inherent challenges of text-only instructions [6]. Several solutions have been proposed by the HCI community to address this issue, such as generating visual mobile tutorials from text-based instructions with HelpViz [9], or creating instructional videos from markdown-formatted texts [1]. Moreover, projects like CAPtAR [8], InstruMentAR [4] and DuploTrack [3] have explored the use of educational augmented reality (AR) overlays. Despite these advancements, a gap persists in seamlessly integrating textual instructions with the user's physical context, which is crucial for ensuring effective task execution.

In this study, we introduce our ongoing work InstructAR, a novel tool designed to translate textual instructions into AR guides that directly align with real-world actions. InstructAR leverages technologies, including Natural Language Processing (NLP) and Computer Vision (CV), to enhance in-situ instructional content delivery. With the power of NLP, InstructAR can parse and interpret text inputs, while CV enables real-time object tracking within the user's environment. This fusion of technologies provides users with intuitive, context-aware guidance, reducing confusion and minimizing mistakes.

2 OBSERVING MANUAL TASK VIDEOS

To inform our initial design, we began by reviewing relevant videos from the large-scale Ego4D dataset [2], and observing how people handle instructional manual tasks. Additionally, we analyzed how-to videos and their corresponding manuals available online. We identified three key issues: **(1) Object Identification:** When multiple objects are involved, it is often unclear which object to focus on. **(2) Action Specification:** Manuals frequently provide high-level descriptions that lack specific, actionable details. For example, instructions like "Secure the shelf to the wall" are vague about the exact steps required. **(3) Step Validation:** There is no immediate feedback mechanism to confirm whether a step has been correctly executed, making it easy for mistakes to propagate and disrupt subsequent steps.

InstructAR aims to address these challenges by converting textual instructions into intuitive, visual AR guides that align with real-world actions. Our initial set of action shapes is inspired by tutorials from simulator games to ensure comprehensibility. InstructAR parses high-level steps into low-level actionable steps, identifies objects in focus, and provides feedback on the correctness of each step. The system also employs progression logic to update relevant guides for the next step.

By leveraging the immersive nature of AR technology, InstructAR provides users with interactive, contextually relevant guidance directly within their physical environment, bridging the gap between abstract instructions and concrete, actionable steps.

3 SYSTEM OVERVIEW

InstructAR operates through a series of key functionalities designed to facilitate the seamless translation of textual instructions into immersive AR guides:

1. Translation of Textual Instructions into AR Guides: The process begins with the capturing and parsing of high-level textual instructions to identify individual actions, relevant objects, and

their sequences using string matching. These instructions are then mapped to detailed, actionable steps, ensuring the specificity required for AR guide generation. For each low-level step, a JSON object is created, encompassing fields such as `mainObject`, `operation`, `destination`, `postCondition` and `textIndex`, which serve as the blueprint for AR guide generation. Once the parsing is complete, the system kicks off the task by starting at step 1 of the instruction.

2. Real-Time Object Tracking: InstructAR integrates with ReactJS and tracking.js [5], a JavaScript library for live color-based object tracking. This enables InstructAR to identify and anchor AR guides to relevant physical objects, enhancing the contextual relevance of the instructional content based on the parsed JSON. The object detection mechanism continuously scans the environment to identify objects based on predefined color signatures.

3. Positive Reinforcement and State Progression: As a result of #1, InstructAR evaluates the current state against the `postCondition` of each step. If it confirms the completion of the step, it progresses to the next low-level instruction. In the current system, the conditions are defined based on object locations. When moving to a new high-level step, InstructAR updates the UI with green label on the top to show success (Fig. 1). After pausing for 2 seconds, the system updates the text progresses to the next step, ensuring that users are presented with relevant guidance at the appropriate stage of the task.

4. Dynamic UI Updates: The UI update consists of three parts. First, InstructAR draws bounding boxes around the `mainObject` and, if not null, the `destination` object. Second, the system generates shapes to guide the action based on the operation of that step. Lastly, InstructAR updates the upper label text according to the `textIndex`, mapping back to the original textual input. These updates occur every time tracking.js detects a change from the camera.

4 FUTURE WORK

As mentioned before, this project is still in its infancy. More formal research is needed to inform and iterate our design. Regarding system development, we will focus on three key areas to handle more complex tasks: **(1) Improving CV Capabilities:** Enhancing the system to support a broader range of operations beyond just comparing object location in the backend. **(2) Implementing Reversible Actions:** Enabling the system to detect user errors and provide suggestions for mitigation. **(3) Migrating to HMD:** Running the system on a Head-Mounted Display (HMD) for a more natural user experience compared to a web-based setup with a webcam. These improvements will enhance the stability, versatility, and usability of InstructAR. Once completed, we expect a better system design, and the prototype should be ready for testing against other traditional methods.

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